Lab 4 Worksheet - Practice

Q1. Extract and clean text patterns from strings using regular expressions.
You are given a messy dataset of energy sources with inconsistent units and spacing.

```
energy_sources <- c("Solar (MWh)", "Wind (GWh)", "Coal (MWh)", "Hydro (kWh)")
```

a. Use str_detect() to identify all energy sources that contain the substring "Wind" or "Hydro" (hint: use the | operator in regex).
b. Extract the unit type (e.g., MWh, GWh, kWh) using str_extract(). Make sure your pattern handles extra spaces or hyphens.
c. Clean the names to keep only the energy source name, trimming whitespace and removing parentheses.
d. Create a clean tibble called energy_clean with two columns: source and unit.
e. Discussion: Why might regex cleaning be important before performing joins or aggregations?

Q2. Joining State Datasets: Socioeconomic + Regional Data
We'll use built-in datasets to explore US state-level data. state.x77 contains metrics such as population, income, and life expectancy. state.region classifies each state into one of four regions: Northeast, South, North Central, West.

```
# Load datasets
library(dplyr)
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
# Convert matrix to data frame
states <- data.frame(state.x77)

# Add state names as a new column
states$state <- rownames(states)

# select and reorder columns
states <- states |>
  select(state, Income, Population, Illiteracy)
# table for regions
regions <- data.frame(state = state.name, region = state.region)
```

a. Perform a left join to combine socioeconomic and regional data.
b. Perform an inner join. How many rows remain, and why is it the same or different?
c. Create a summary table that shows average Income and average Life Exp by region.
d. Perform an anti join to check if any state names don't match between datasets. What might cause mismatches like this in real projects?
e. Discussion: Why would you prefer a left join over an inner join when merging state-level datasets?

f. Use pivot_longer() to make the socioeconomic data longer — convert the numeric columns into key-value pairs.

Q3. Mapping
Let's visualize data by state.

```r
library(ggplot2)
library(maps)
library(dplyr)

us_map <- map_data("state")

state_data <- states |>
  left_join(regions, by = "state") |>
  mutate(state = tolower(state))
```

Use a join to combine the us_map with the state_data

```r
us_map_joined <- ...
```

Create a chloropleth map:

```r
ggplot(us_map_joined, aes(long, lat, group = group, fill = Income)) +
  geom_polygon(color = "white") +
  coord_fixed(1.3) +
  theme_minimal() +
  scale_fill_viridis_c() +
  labs(title = "Average Income by State")
```

b. Swap Income for Life Exp to visualize life expectancy instead. Try changing the color scale using scale_fill_gradient() or scale_fill_gradient2()

Q4. Quarto Dashboard Setup
Create a new file ev-dashboard.qmd and add:

```
---
title: "EV Power Dashboard"
format: dashboard
---

Add some sections:

### Overview

Your project introduction.

### Map Visualization

# Paste your ggplot map code here

### Data Table

# Show your joined dataset

head(us_map_joined)

Then render:

quarto preview
```